

Team Collaboration & Integration Guide

Insurance Policy & Claims Management System | Architecture & Workflow

1. Project Architecture & Tech Stack

Backend: Python 3.12, FastAPI, Uvicorn, SQLAlchemy ORM, SQLite (insurance.db), Pydantic v2, PyJWT, Bcrypt.

Frontend: ReactJS 18, Material UI (MUI v5), Recharts, Axios, React Router v6, Vite.

Repository: <https://github.com/harshithcheripally16-ui/Insurance-Policy-Claims-Management.git>

2. Module Assignments & Responsibilities

Module	Role	Key Features & Pages	Status
Insurance Agent	Agent	Agent Portfolio, Sales (📊), Client Policies, Customer Directory	Completed (main)
Customer Module	Customer	Browse Policy Catalog, Purchase Plans, Submit Claims & Track Reviews	Open for Integration
Claims Officer	Claims Officer	Review Workbench, Risk Score Engine (0-100), Document Verification	Open for Integration
Admin Module	System Admin	Master Financial Analytics, User Directory, Policy Catalog Creation	Open for Integration

3. Initial Developer Setup Commands

Step 1: Clone Repository & Create Feature Branch

```
git clone https://github.com/harshithcheripally16-ui/Insurance-Policy-Claims-Management.git
cd Insurance-Policy-Claims-Management
git checkout -b feature/customer-module # (or feature/claims-officer-module / feature/admin-module)
```

Step 2: Set Up Backend (Python FastAPI)

```
cd backend
python -m venv venv
.\venv\Scripts\activate # (macOS/Linux: source venv/bin/activate)
pip install -r requirements.txt
python -m app.seed
python run.py # Running at http://127.0.0.1:8000 (Swagger docs: http://127.0.0.1:8000/docs)
```

Step 3: Set Up Frontend (ReactJS + Vite)

```
cd frontend
npm install
npm run dev # Running at http://localhost:3000
```

4. Back-End Integration Instructions (backend/app/)

Place all your API endpoints inside your assigned router file:

- Customer: backend/app/routers/customer_router.py
- Claims Officer: backend/app/routers/claims_officer_router.py
- Admin: backend/app/routers/admin_router.py

Router File Example (backend/app/routers/your_module_router.py):

```
from fastapi import APIRouter, Depends
from sqlalchemy.orm import Session
from app.database import get_db
from app.models import User, UserRole
from app.dependencies import require_roles
```

```

router = APIRouter(prefix="/api/your-module", tags=["Your Module"])\

@router.get("/dashboard")
def get_data(db: Session = Depends(get_db), user: User = Depends(require_roles([UserRole.CUSTOMER]))):
    return {"message": "Module loaded"}

```

Register Router in backend/app/main.py:

```

from app.routers import your_module_router
app.include_router(your_module_router.router)

```

5. Front-End Integration Instructions (frontend/src/)

Create your page in frontend/src/pages/ and register it in frontend/src/App.jsx:

```

// In frontend/src/App.jsx
import CustomerDashboard from './pages/CustomerDashboard';

<Routes>
  <Route path="/customer/dashboard" element={<CustomerDashboard />} />
</Routes>

```

6. Pre-Seeded Test Credentials

Role	Email	Password	Module Developer
■ Insurance Agent	agent@insure.com	password123	Completed (Priya Nair)
■ Customer	customer@insure.com	password123	Customer Module Teammate
■ Claims Officer	officer@insure.com	password123	Claims Officer Teammate
■ Admin	admin@insure.com	password123	Admin Teammate

7. Git Commit & Push Guidelines

```

git status
git add .
git commit -m "feat: implement <your-module-name> functionality"
git push origin feature/<your-module-name>
# Open Pull Request (PR) on GitHub to merge into main.

```